

EXPERIMENT NO. 1

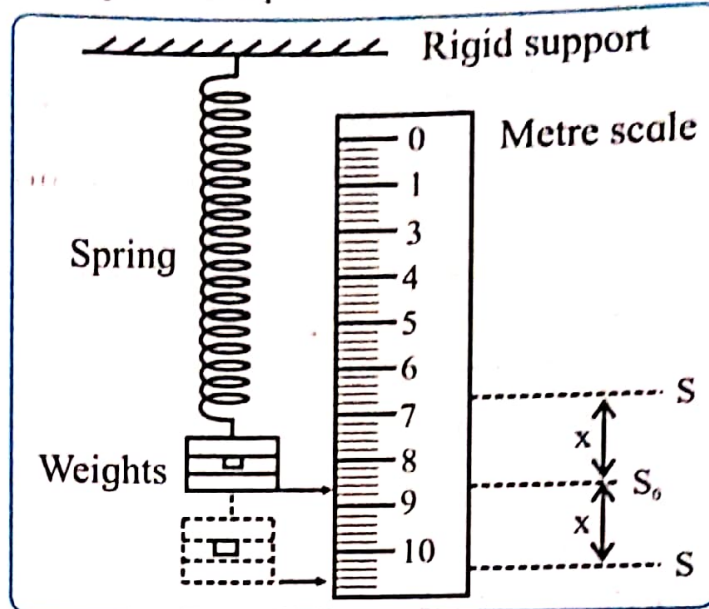
SPRING - MASS OSCILLATOR

Aim : a) To determine the force constant (k) of the given spring.

b) To determine the mass of the spring.

Apparatus : Light spiral spring with clamping arrangement and attached pointer, meter scale, hanger, weight box, stop watch .

Diagram :



Formula :

1) $F = -kx$ 2) Potention energy (P.E) $= \frac{1}{2} Kx^2$ 3) mass of spring $m_s = 3 \times (x\text{-intercept})$

Where, F - is restoring force.

x - is extension of the spring.

k - is force constant of the spring.

Theory:

The period (T) of S.H.M. about the position of equilibrium

$$T^2 = \frac{16\pi^2 R^3 N}{r^4 n} \left[M = \frac{m_s}{3} \right]$$

differentiating above equation, we get

$$T^2 = 0,$$

$$0 = \frac{16\pi^2 R^3 N}{r^4 n} \left[M = \frac{m_s}{3} \right]$$

$$\therefore M + \frac{m_s}{3} = 0$$

$$\therefore M = -\frac{m_s}{3}$$

$$\therefore m_s = 3 \times (x \text{ intercept})$$

$$\frac{m_s}{3} = x \text{ intercept}$$

$$\therefore m_s = 3x (x \text{ intercept})$$

Where,

- R - Radius of spring
N - No. of turns in the spring
r - Radius of the wire used for spring
n - Rigidity of the material of the spring
T - Time period of oscillation
Ref : Advanced Practical Physics for students by B. L. Warnop & Flint

Procedure :

PART - A To find force constant (K) & (P.E)

- 1) Weigh the hanger and determine the mass of the hanger (m_0).
- 2) Clamp the given spiral spring to a rigid support and attach the hanger with pointer at its lower end.
- 3) Add a suitable mass to the hanger so that the spring is stretched to about triple of the unstretched length. i.e each turn of the spring will be free from each other.
- 4) Note down total mass attached to the spring (M) and also note down the reading (S_0) i. e. mean position on the scale according to position of the pointer.
- 5) Add one slotted weight of 50g in the hanger and note the position of the pointer say S. Repeat the procedure twice by adding weights in steps of 50g. (m is more than M.)
- 6) Remove weights from the hanger and bring the pointer to its mean position S_0 . Take three readings by reducing weights in steps of 50g. (m less than M)
- 7) Determine the extension ($S_0 - S$) = x in each case.
- 8) Plot the graph of F against x and determine slope of the graph. Slope of the graph is force constant (k)
- 9) Calculate the potential energy $\left(\frac{1}{2}kx^2\right)$ for each value of extension x and plot the graph of potential energy against extension x.

PART - B To find mass of the spring m_s

- 1) Put suitable weights in the hanger and determine the time to complete 20 oscillations by using stop watch. Repeat this step two times, hence calculate mean time (t) for 20 oscillations.
- 2) Calculate the periodic time (T).
- 3) Note down the next four observations by changing the mass in steps of 50g. Find the time for 20 oscillations in each case. Where, total mass M (mass of the hanger + mass kept in the hanger).
- 3) Determine mean time (t) in each case. Calculate periodic time (T) for each mass.
- 4) Plot the graph of T^2 against M and hence find the mass of the spring m_s

Observations :

PART - A To determine force constant (k) and Potential Energy (P.E.):

1. Mass of the hanger (m_0) = .59.. g
2. Mass attached to the hanger = m_1 = .159. g, so that spring gets stretched to about triple of its unstretched length. Therefore total mass attached $M = (m_1 + m_0) = 209$ g.
3. Position of the pointer when the spring is stretched to about triple of the unstretched length S_0 = .29.5.... cm.

4. To determine force constant (k) of the spring:

Sr. No.	Mass attached to the spring m gwt	F = (m-M) gwt	S cm	Extension $S_0 - S = x$ cm	P.E. = $\frac{1}{2} k x^2$
1	50	-150	16.7	8.8	645.46
2	100	-100	19.2	6.3	330.81
3	150	-50	22.5	3.0	75.01
4	M = 200	0	$S_0 = 25.5$	0	0
5	250	+50	28.8	-3.3	90.76
6	300	+100	32.0	-6.5	392.15
7	350	+150	34.7	-9.2	705.47

Observations :

PART - B To determine the mass of spring (m_s):

- Mass of the hanger (m_0) = ...52... g
- Least count of the stop watch = ...15... s

Sr No.	Total Mass M kg	Time t for 20 oscillation (s)			mean t (s)	Periodic time T = mean t/20 (s)	T ² (S ²)
		1	2	3			
1	50	8	8	7	7.6	0.38	0.1444
2	100	12	10	11	11	0.55	0.3025
3	150	14	13	13	13.3	0.665	0.4422
4	200	15	15	15	15	0.75	0.5625
5	250	17	16	17	16.6	0.83	0.6889

Calculations:

1) force constant K of the spring = slope of graph of F against extension X
 $= \dots 16.67 \dots \text{gwt/cm.}$
 $= 163.33 \dots \text{dyne/cm.}$
 $= 1.63366 \times 10^4 \text{ dyne/cm}$

2) potential energy = $\frac{1}{2} Kx^2 \Rightarrow \frac{1}{2} (16.67) (77.44) = 645.46$

$\Rightarrow \frac{1}{2} (16.67) (39.69) = 330.81$

$\Rightarrow \frac{1}{2} (16.67) (9) = 75.01$

$\Rightarrow \frac{1}{2} (16.67) (0) = 0$

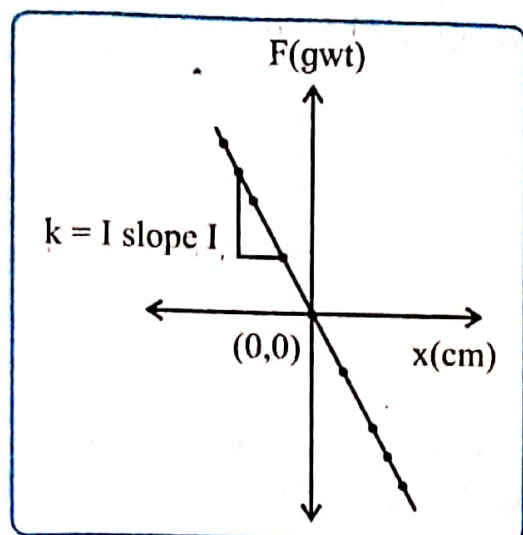
$\Rightarrow \frac{1}{2} (16.67) (10.89) = 90.76$

$\Rightarrow \frac{1}{2} (16.67) (42.25) = 352.15$

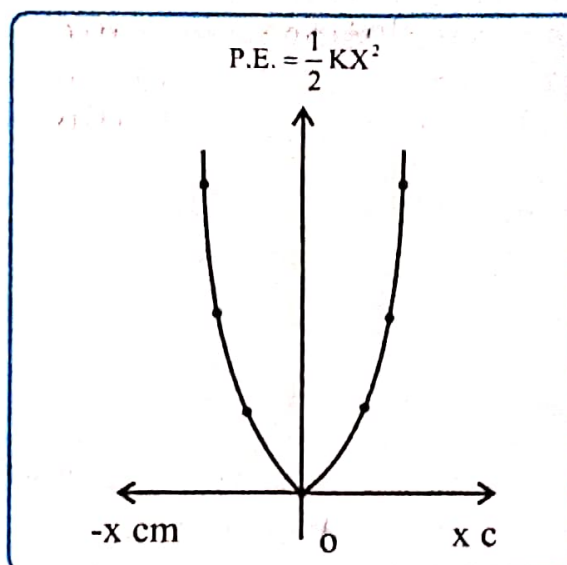
$\Rightarrow \frac{1}{2} (16.67) (84.64) = 705.47$

Graphs:

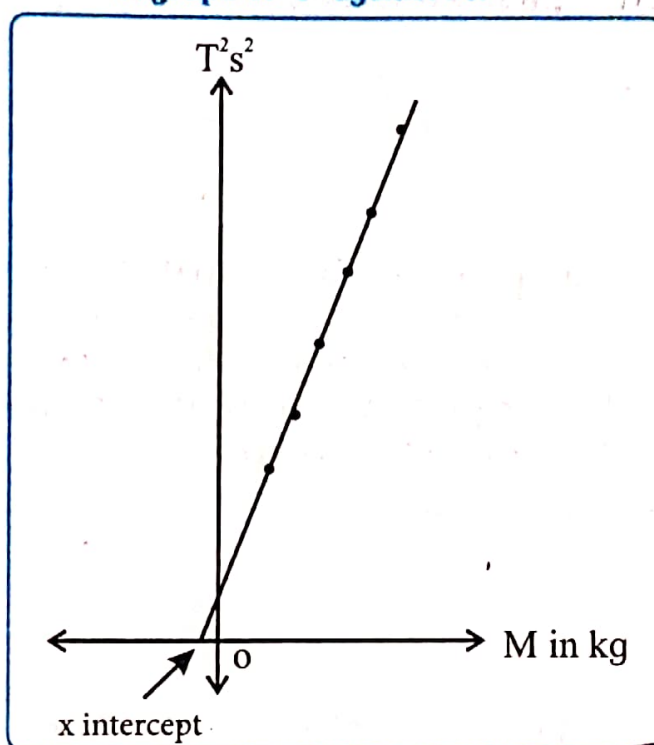
A graph of F against x:



A graph of P.E. against x:



A graph of T^2 against M:



Result :

1. Force constant of the given spring $(k) = \dots\dots\dots$ dyne / cm
2. Mass of the given spring (m_s) from the graph of T^2 against $M = \dots\dots\dots$ g
3. The graph shows that potential energy of oscillator changes with distance x from equilibrium. The nature of graph is parabolic.

Precautions :

1. Record the mean position of the pointer carefully.
2. Oscillations of the spring mass oscillator should be in a vertical plane.
3. The pointer should move freely over the scale such that it should not touch the scale.
4. Spiral spring should not be stretched beyond the elastic limit.

Questions

1. What are damped oscillations?

The oscillations in which the amplitude gradually decreases with time due to dissipative forces are called damped oscillations.

2. Define linear S. H. M.?

Linear Simple Harmonic Motion is the periodic motion in which the restoring force (or acceleration) is directly proportional to the displacement from the mean position and is always directed towards the mean position.

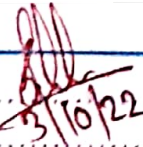
3. What are forced oscillations?

The oscillations of a body under the influence of an external periodic force are called forced oscillations.

4. Define force constant.

Force constant is the restoring force produced per unit displacement of a body from its mean position. It is denoted by k .

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(A7)

Y ↑ F (gwt)

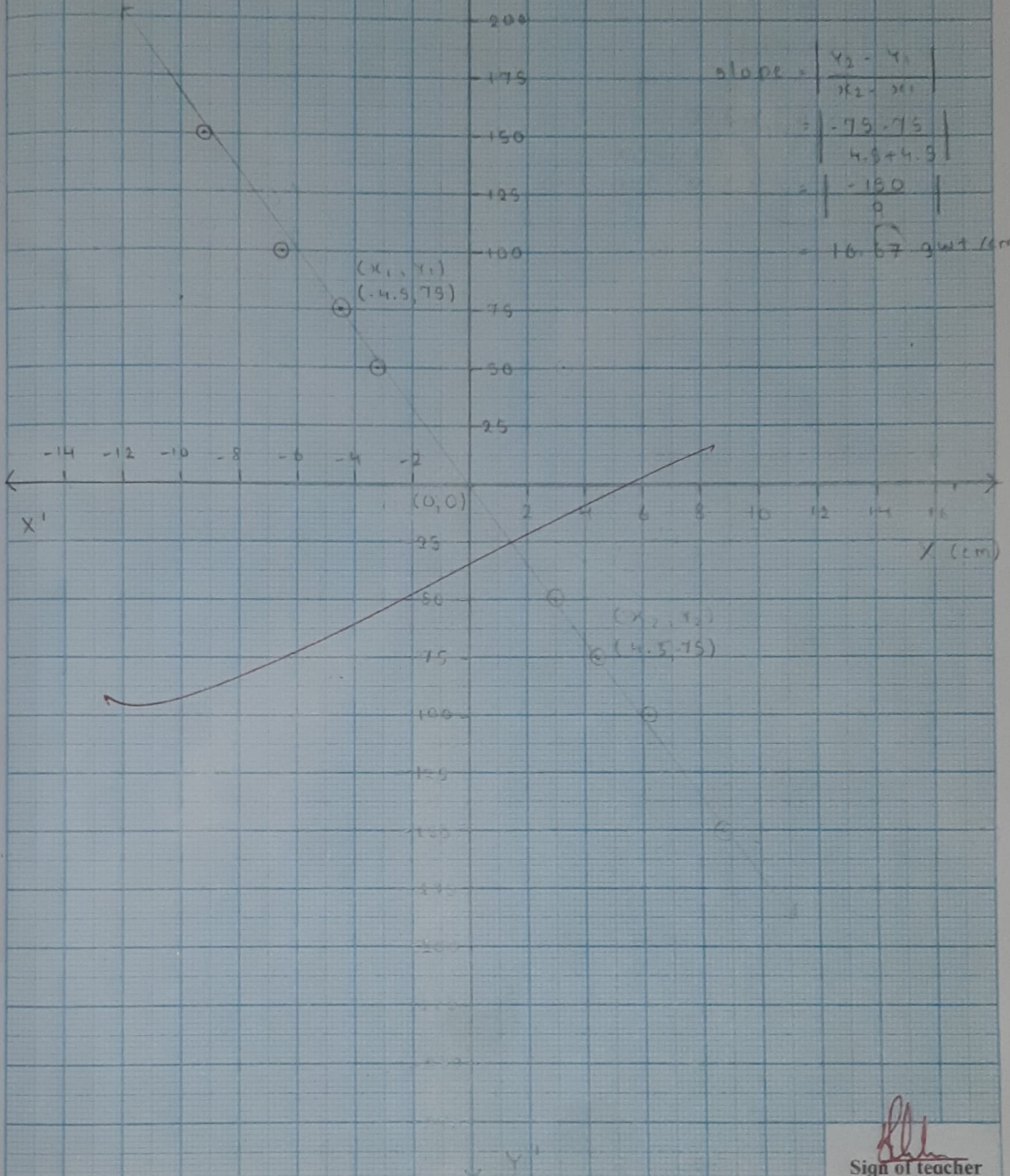
Scale :

Scale

X axis : 1 unit = 2 cm

Y axis : 1 unit = 25 gwt

$$\begin{aligned} \text{slope} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-75 - 75}{4.5 + 4.5} \\ &= \frac{-150}{9} \\ &= -16.67 \text{ gwt/cm} \end{aligned}$$



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NOTES / CALCULATIONS / ROUGH WORK

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-95 - 75}{4.5 - (-4.5)}$$

$$= \frac{-150}{9}$$

$$= -16.67$$

$$= 16.67 \text{ gwt/cm}$$

$$= 16.67 \times 980 \text{ dyne/cm}$$

$$= 16336.6 \text{ dyne/cm}$$

$$\text{slope} = 1.63366 \times 10^4 \text{ dyne/cm}$$

$$PE = \frac{1}{2} k x^2$$

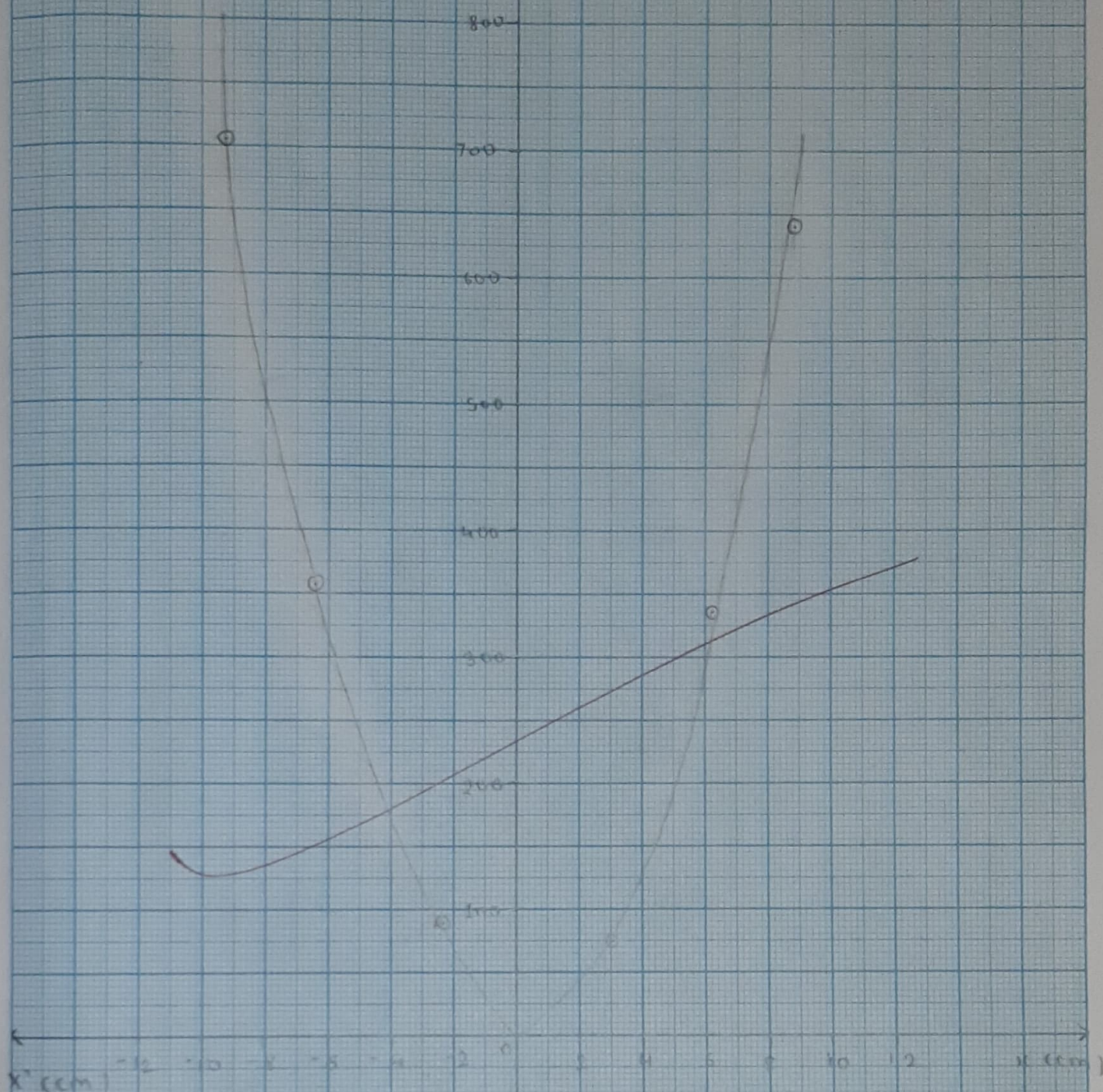
Scale :

x-axis

1 cm = 2 mm

y-axis

1 cm = 50



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$T^2 s^2 \uparrow$

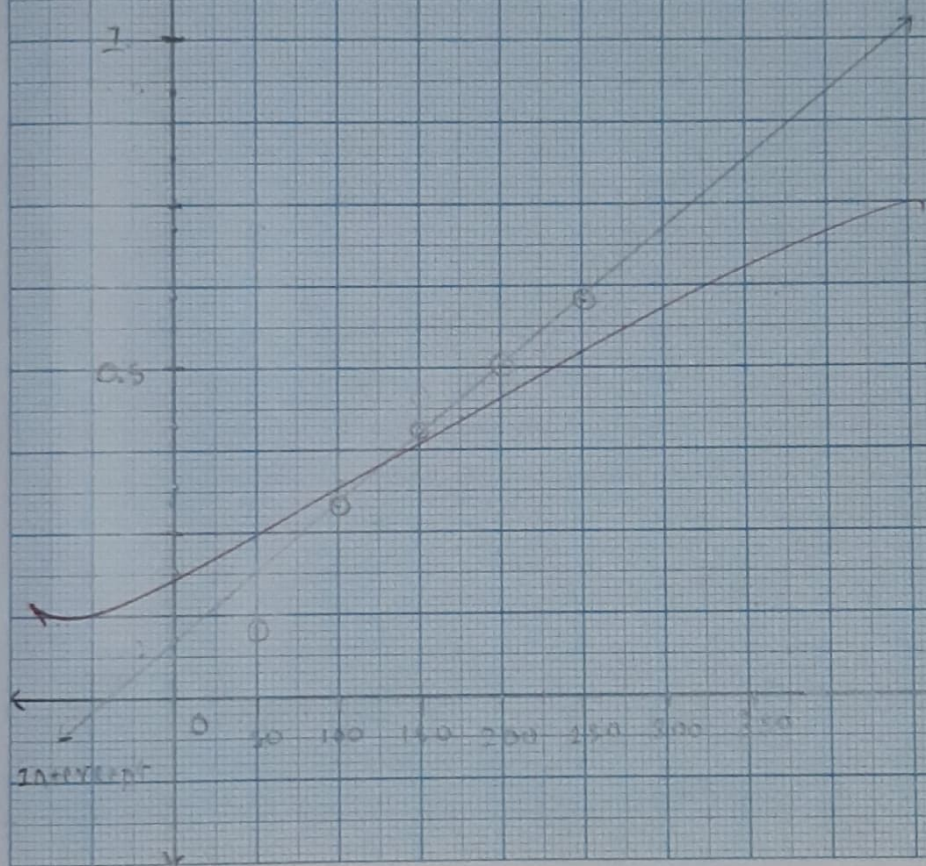
Scale :

X - axis

1 cm = 50 g

Y - axis

1 cm = 1 s²



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